IMAT 모의고사 7회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Telemedicine expanded rapidly when clinics closed their doors during the pandemic, and many health systems have kept remote consultations since. Supporters argue that video visits save patients travel time and reduce missed appointments, particularly in rural areas. Opponents warn that physical examination is impossible on a screen and that patients without reliable internet or digital skills — often older or poorer — may be left behind. A sensible policy, some suggest, would offer remote care as an option rather than a replacement.

1. Which of the following best summarises the opponents' argument?

A. Telemedicine was invented during the pandemic.
B. Missed appointments increase with telemedicine.
C. Telemedicine is more expensive than in-person care.
D. Remote care has limits and may worsen inequalities in access.
E. Rural patients prefer in-person visits.

2. The phrase "an option rather than a replacement" implies that:

A. in-person care should remain available alongside remote care.
B. all consultations should be remote.
C. internet access should be mandatory.
D. remote care should be abolished.
E. only rural patients should use remote care.

3. Which Italian city hosts the oldest university in continuous operation in the Western world, founded in 1088?

A. Rome
B. Naples
C. Bologna
D. Pisa
E. Padua

4. The Paris Agreement (2015) is an international treaty concerned mainly with:

A. nuclear disarmament
B. climate change
C. refugee rights
D. international trade tariffs
E. public health emergencies

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Doctors who work in emergency departments are trained in resuscitation. Dr Rossi is trained in resuscitation." What can be concluded?

A. Dr Rossi is not a doctor.
B. All doctors trained in resuscitation work in emergency departments.
C. Dr Rossi works in an emergency department.
D. Dr Rossi might or might not work in an emergency department.
E. Dr Rossi does not work in an emergency department.

6. A laboratory sample contains 2400 bacteria. The population doubles every 30 minutes. How many bacteria are there after 2 hours?

A. 38 400
B. 9600
C. 19 200
D. 4800
E. 76 800

7. Six seats are arranged in a row. A sits at one end. B sits next to A. C sits at the other end. D does not sit next to C. E and F sit next to each other. Which seat (1–6, with A in seat 1) does D occupy?

A. 2
B. 5
C. 4
D. Cannot be determined
E. 3

8. What is the next term in the sequence: 1, 4, 9, 16, 25, __?

A. 49
B. 36
C. 35
D. 32
E. 30

9. A recipe for 4 people uses 300 g of rice. How much rice is needed for 10 people?

A. 700 g
B. 750 g
C. 650 g
D. 1200 g
E. 600 g

Section 3 — Biology (Q10–32)

10. Which of the following is the function of the nucleolus?

A. Protein degradation
B. Synthesis of lipids
C. Assembly of ribosomal subunits and rRNA synthesis
D. ATP production
E. DNA replication only

11. Which of the following is required for transcription in eukaryotes?
A. RNA polymerase binding to a promoter
B. Okazaki fragments
C. tRNA carrying amino acids
D. Ribosomes binding to DNA
E. DNA polymerase

12. Cancer cells commonly show:
A. absence of a nucleus.
B. uncontrolled division due to mutations in proto-oncogenes and tumour suppressor genes.
C. contact inhibition.
D. inability to replicate DNA.
E. limited division.

13. In oxidative phosphorylation, ATP synthase is driven by:
A. direct oxidation of oxygen.
B. glucose diffusion.
C. a gradient of Na^+ ions.
D. ATP hydrolysis.
E. a proton (H^+) gradient across the inner mitochondrial membrane.

14. In a family, a disease appears in every generation, affected fathers pass it to all daughters but no sons. The most likely inheritance is:
A. autosomal dominant.
B. X-linked dominant.
C. X-linked recessive.
D. autosomal recessive.
E. mitochondrial.

15. Which of the following organisms are primary producers in most ecosystems?
A. Herbivores
B. Fungi
C. Decomposers
D. Photosynthetic organisms
E. Carnivores

16. Surfactant in the alveoli functions to:
A. increase surface tension.
B. kill bacteria.
C. produce mucus.
D. reduce surface tension and prevent alveolar collapse.
E. transport oxygen.

17. Which of the following is a disaccharide composed of glucose and galactose?
A. Fructose
B. Cellulose
C. Lactose
D. Sucrose
E. Maltose

18. The refractory period of a neuron ensures that:
A. the axon becomes myelinated.
B. the neuron fires continuously.
C. K^+ ions remain inside.
D. action potentials travel in one direction.
E. neurotransmitters are reabsorbed.

19. Which of the following vessels has the thinnest walls, allowing exchange of substances?
A. Arterioles
B. Aorta
C. Arteries
D. Veins
E. Capillaries

20. In cats, the gene for orange coat (X^O) and black coat (X^B) are X-linked and codominant. A tortoiseshell (orange-and-black) cat is almost always:
A. female.
B. either sex equally.
C. male.
D. sterile.
E. homozygous.

21. Negative feedback in hormone regulation means:
A. the hormone stimulates its own further release.
B. the effect of the hormone reduces further secretion of that hormone.
C. hormone levels rise without limit.
D. the hormone is secreted only at night.
E. hormones are always inhibited.

22. If one strand of DNA has the sequence 5'-ATGCTA-3', the complementary strand is:
A. 3'-TACGAT-5'
B. 5'-ATGCTA-3'
C. 3'-ATGCTA-5'
D. 5'-TACGAT-3'
E. 3'-UACGAU-5'

23. Which cells present antigens to helper T cells using MHC class II molecules?
A. All nucleated cells only
B. Dendritic cells, macrophages and B cells
C. Erythrocytes
D. Platelets
E. Neurons

24. The juxtaglomerular apparatus releases renin when:
A. ADH levels increase.
B. blood glucose rises.
C. blood pressure falls.
D. urine is concentrated.
E. blood pressure rises.

25. Which of the following is a source of genetic variation in sexually reproducing organisms?
A. Vegetative propagation
B. Cloning
C. Binary fission
D. Mitosis
E. Independent assortment and crossing over

26. Which bond is responsible for the tertiary structure linking cysteine residues?
A. Hydrogen bond
B. Glycosidic bond
C. Disulfide bond
D. Peptide bond
E. Phosphodiester bond

27. A karyotype shows 47,XXY. This condition is called:
A. Klinefelter syndrome
B. Fragile X syndrome
C. Edwards syndrome
D. Turner syndrome
E. Down syndrome

28. Which of the following crosses the plasma membrane most easily by simple diffusion?
A. Proteins
B. O_2
C. Na^+
D. Glucose
E. Cl^-

29. Prions are infectious agents made of:
A. lipids
B. carbohydrates
C. misfolded proteins
D. DNA only
E. RNA only

30. Where is most water absorbed in the digestive tract?
A. Mouth
B. Stomach
C. Small intestine
D. Oesophagus
E. Rectum only

31. Which of the following is true of prokaryotic gene expression?
A. Ribosomes are 80S.
B. Introns are always present.
C. mRNA must be exported from a nucleus.
D. A 5' cap is added to mRNA.
E. Transcription and translation can occur simultaneously.

32. A species of moth darkened after soot polluted tree bark, then lightened after air quality improved. This illustrates:

A. genetic drift.
B. Lamarckism.
C. spontaneous generation.
D. directional natural selection.
E. artificial selection.

Section 4 — Chemistry (Q33–47)

33. What is the mass of 3×10^{23} molecules of water? (H_2O, M = 18 g/mol, N_A = 6×10^{23})

A. 6 g
B. 9 g
C. 18 g
D. 3 g
E. 36 g

34. In which of the following is the oxidation number of carbon +4?

A. CH_3OH
B. CH_4
C. CO_2
D. C_2H_6
E. CO

35. At 25 °C, a solution has pH 3. What is $[OH^-]$?

A. 10^{-14} mol/L
B. 10^{-3} mol/L
C. 10^{3} mol/L
D. 10^{-11} mol/L
E. 10^{-7} mol/L

36. Which of the following atoms has the smallest atomic radius?

A. C
B. Be
C. B
D. Li
E. F

37. Which of the following has a linear shape?

A. CH_4
B. SO_2
C. H_2O
D. $BeCl_2$
E. NH_3

38. How many grams of CO_2 are produced when 12 g of carbon burns completely in oxygen? (C = 12, O = 16)

A. 88 g
B. 12 g
C. 44 g
D. 22 g
E. 32 g

39. Which reagent distinguishes an aldehyde from a ketone by forming a silver mirror?

A. Tollens' reagent
B. Limewater
C. Universal indicator
D. Sodium chloride
E. Bromine water

40. What is the mass percentage of a solution prepared by dissolving 20 g of salt in 80 g of water?

A. 16%
B. 20%
C. 40%
D. 80%
E. 25%

41. Which statement about metallic bonding is correct?

A. Metallic bonds form between non-metals.
B. Positive ions are held in a sea of delocalised electrons.
C. Metals do not conduct electricity.
D. Electrons are fixed between two atoms.
E. Metals are always liquids.

42. Which pair is a conjugate acid–base pair?

A. NH_4^+ / NH_3
B. H_2O / H_2
C. HCl / NaOH
D. H_2SO_4 / SO_4^{2-}
E. HNO_3 / NO_2

43. Le Chatelier's principle predicts that adding more reactant to a system at equilibrium will:

A. stop the reaction.
B. shift the equilibrium towards the reactants.
C. change the equilibrium constant.
D. have no effect.
E. shift the equilibrium towards the products.

44. What is the general formula of alkanes?

A. C_nH_n
B. C_nH_{2n}
C. C_nH_{2n+2}
D. C_nH_{2n-2}
E. $C_nH_{2n+1}OH$

45. How many unpaired electrons does a ground-state nitrogen atom have?
A. 1
B. 0
C. 2
D. 5
E. 3

46. Which of the following is a reducing agent in the reaction $CuO + H_2 \rightarrow Cu + H_2O$?
A. H_2
B. H_2O
C. Cu
D. CuO
E. O^{2-}

47. Polymerisation of ethene produces:
A. polystyrene
B. nylon
C. PVC
D. polyester
E. polyethene (polyethylene)

Section 5 — Physics & Mathematics (Q48–60)

48. A body starts from rest with acceleration 2 m/s². What distance does it cover in the 3rd second only (between t = 2 s and t = 3 s)?
A. 2 m
B. 9 m
C. 4 m
D. 5 m
E. 3 m

49. A 2 kg object falls from a height of 5 m. What is its speed just before hitting the ground? (g = 10 m/s², no air resistance)
A. 5 m/s
B. 50 m/s
C. 7 m/s
D. 100 m/s
E. 10 m/s

50. What is the resistance of a lamp rated 240 V, 60 W?
A. 240 Ω
B. 14 400 Ω
C. 4 Ω
D. 0.25 Ω
E. 960 Ω

51. An object is placed 30 cm in front of a converging lens of focal length 10 cm. Where is the image formed?

A. 15 cm from the lens on the opposite side
B. No image is formed
C. 7.5 cm from the lens, same side as object
D. 20 cm from the lens on the opposite side
E. 30 cm from the lens on the opposite side

52. A gas at 2 atm occupies 3 L at 300 K. What volume does it occupy at 1 atm and 600 K?

A. 9 L
B. 6 L
C. 12 L
D. 1.5 L
E. 3 L

53. Two forces of 3 N and 4 N act on an object at right angles. What is the magnitude of the resultant force?

A. 5 N
B. 3.5 N
C. 7 N
D. 1 N
E. 12 N

54. An electric kettle of power 2000 W heats water for 3 minutes. How much energy is supplied?

A. 36 000 J
B. 360 000 J
C. 6 000 000 J
D. 6000 J
E. 120 000 J

55. Solve: $4^x = 8$.

A. 2/3
B. 1
C. 3
D. 3/2
E. 2

56. What is the value of $\tan(45°) + \sin(90°)$?

A. 2
B. $1 + \sqrt{2}/2$
C. 0
D. $\sqrt{2}$
E. 1

57. In a class, 40% of students are boys. 50% of boys and 25% of girls wear glasses. What percentage of the class wears glasses?

A. 75%
B. 30%
C. 40%
D. 35%
E. 37.5%

58. What is the area of a triangle with vertices (0, 0), (6, 0) and (0, 4)?

A. 24
B. 10
C. 20
D. 48
E. 12

59. What is the derivative of $f(x) = 3x^2 + 2x$?

A. $6x^2 + 2$
B. $3x + 2$
C. $x^3 + x^2$
D. $6x + 2$
E. $6x$

60. An X-ray photon has energy $E = hf$. If the frequency is tripled, the energy:

A. is multiplied by $\sqrt{3}$.
B. stays the same.
C. is tripled.
D. is multiplied by 9.
E. is reduced to one third.

정답 및 해설

번호	정답	해설
1	D	신체 진찰 불가 + 디지털 소외층 → 한계와 불평등 심화.
2	A	대면 진료도 계속 유지되어야 함.
3	C	볼로냐 대학(1088).
4	B	기후변화.
5	D	응급실 의사 ⊂ 소생술 훈련자 → 역은 성립 안 함.
6	A	2시간 = 4번 배가 → 2400 × 16 = 38 400.
7	E	A1, B2, C6. D는 5 불가. E·F 인접: 남은 3·4·5 중 D 위치 3이면 E·F는 4·5 (가능); D=4면 E·F 3·5 불인접 → 불가. 따라서 D=3.
8	B	$6^2 = 36$.
9	B	75 g/인 × 10 = 750 g.
10	C	rRNA 합성, 리보솜 소단위 조립.
11	A	RNA 중합효소가 프로모터에 결합.
12	B	원종양유전자·종양억제유전자 돌연변이 → 무제한 분열.
13	E	내막 사이 양성자 기울기(화학삼투).
14	B	X연관 우성: 이환 아버지 → 딸 전부, 아들 없음.
15	D	광합성 생물.
16	D	표면장력↓, 폐포 허탈 방지.
17	C	젖당 = 포도당 + 갈락토스.
18	D	불응기 → 활동전위 한 방향 전도.
19	E	모세혈관.
20	A	X 두 개 필요 → 암컷(XXY 수컷 예외).
21	B	호르몬 효과가 분비 억제.
22	A	상보·역평행: 3'-TACGAT-5'.
23	B	전문 항원제시세포(수지상세포, 대식세포, B세포).
24	C	혈압 저하 시 레닌 분비.
25	E	독립 분리 + 교차.
26	C	이황화 결합.
27	A	클라인펠터 증후군.
28	B	작은 비극성 분자 O_2.
29	C	잘못 접힌 단백질.
30	C	소장.
31	E	핵막 없음 → 전사·번역 동시 진행.
32	D	방향성 선택(공업 암화).
33	B	0.5 mol × 18 = 9 g.
34	C	CO_2: +4.
35	D	pOH 11.

번호	정답	해설
36	E	같은 주기 오른쪽일수록 작음 → F.
37	D	$BeCl_2$ 직선형.
38	C	1 mol C → 1 mol CO_2 = 44 g.
39	A	톨렌스 시약(은거울 반응).
40	B	20/(20+80) = 20%.
41	B	양이온 + 비편재 전자 바다.
42	A	NH_4^+/NH_3는 H^+ 하나 차이.
43	E	반응물 추가 → 정반응 쪽 이동. K는 불변.
44	C	C_nH_{2n+2}.
45	E	$2p^3$: 3개.
46	A	H_2가 산화됨 → 환원제.
47	E	폴리에틸렌.
48	D	s(3) − s(2) = 9 − 4 = 5 m.
49	E	v = √(2gh) = √100 = 10.
50	E	R = V^2/P = 57 600/60 = 960 Ω.
51	A	1/v = 1/10 − 1/30 = 2/30 → v = 15 cm (실상).
52	C	V_2 = 3 × (2/1) × (600/300) = 12 L.
53	A	√(9+16) = 5.
54	B	2000 × 180 = 360 000 J.
55	D	$2^{2x} = 2^3$ → x = 3/2.
56	A	1 + 1 = 2.
57	D	0.4×0.5 + 0.6×0.25 = 0.20 + 0.15 = 35%.
58	E	½ × 6 × 4 = 12.
59	D	6x + 2.
60	C	E ∝ f.